

Multi-Objective Evolutionary Algorithms for Overlapping Community Detection

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Abstract—Community detection in complex networks is a fundamental problem with broad applications across social, biological, and technological domains. Most real-world networks exhibit overlapping community structure, where nodes participate in multiple groups simultaneously. This paper presents an extension of the High-Performance Multi-Objective Community Detection (HP-MOCD) algorithm to support overlapping community assignments through a novel three-objective NSGA-II framework. The extended algorithm introduces a multi-label chromosome representation, topology-aware crossover, and a five-operator mutation system with a two-phase evolutionary strategy that first optimises structural quality and then introduces meaningful overlap. We evaluate our approach against four competing algorithms – MCMOE, CPM-Original, CPM-NCN-Fixed, and SLPA – on two benchmark datasets: the LFR synthetic overlapping benchmark and the DBLP co-authorship network. On the LFR benchmark, our extension recovers 188 of 200 ground-truth overlapping nodes. On DBLP, it detects 5,564 overlapping nodes in a 10,000-node subgraph where 75.5% of nodes belong to multiple communities, aligning closely with real-world co-authorship patterns. Results confirm that multi-objective evolutionary approaches balance detection quality with meaningful overlap recovery in ways that neither structural methods nor label-propagation baselines can consistently achieve.

I. INTRODUCTION

Community detection is fundamental for complex system analysis. It provides crucial insights into the structural organisation of social networks, biological interactions, and citation graphs. Traditionally, this task involves partitioning a network into groups of nodes that are densely connected internally but sparsely connected to the rest of the graph. However, most real-world networks exhibit an overlapping community structure, where individual nodes naturally participate in multiple groups simultaneously, so identifying these overlapping memberships is essential for creating a realistic and expressive model of network dynamics.

The detection of overlapping communities is a significant challenge due to its NP-hard nature and the combinatorial explosion of potential assignments as network size increases. Traditional approaches, such as the Clique Percolation Method (CPM), are often used for detecting overlaps but face critical limitations, including high computational costs and the non-classified node (NCN) problem, where nodes not part of a specific clique remain unassigned. Furthermore, single-objective optimisation approaches often fail to capture the complexity of community structures, as they are confined to

optimising a single function that may not account for the conflicting goals of maximising intra-community density while minimising inter-community links.

To address these challenges, Multi-objective Optimisation Evolutionary Algorithms (MOEAs) are uniquely suited because they explore multiple competing solutions simultaneously, producing a set of Pareto-optimal community structures. However, standard MOEAs often struggle with efficient representation schemes for overlapping nodes, frequently requiring complex and computationally expensive decoding processes. Recent advancements have suggested that utilising maximal-clique graphs can simplify this optimisation by making overlap an intrinsic property of the representation.

This paper proposes an extension of the High-Performance Multi-Objective Community Detection (HP-MOCD) algorithm to support expressive community modelling through overlapping memberships. Our approach aims to combine the scalability of the HP-MOCD framework with advanced structural techniques, such as clique-based representations and modularity optimisation, to ensure both high partition accuracy and complete node coverage. We evaluate the performance of our extended algorithm using the LFR Benchmark Network for controlled quantitative analysis and the DBLP Collaboration Network to assess practical effectiveness on real-world data. Performance is measured using standard metrics, including Modularity, Normalised Mutual Information (NMI), Adjusted Mutual Information (AMI), pairwise F1, Omega Index, and Overlapping NMI (ONMI).

II. RELATED WORK

1. HP-MOCD (2025) [1]

Santos et al. propose HP-MOCD, a scalable MOEA for community detection that introduces efficient genetic operators and elitist NSGA-II selection to reduce fitness evaluation overhead. By simultaneously optimising intra-community density and size balance, it produces a well-distributed Pareto front with runtime complexity $O(G \cdot N_p|V|)$. Its primary limitation is that the representation restricts each node to a single community, making it the foundational disjoint baseline our work extends.

2. Overlapping Community Detection Survey (2013) [2]

Xie et al. survey major overlapping detection paradigms – clique percolation, label propagation, link partitioning, and

statistical inference – and introduce a unified evaluation framework using NMI, Omega Index, and F-score. The study finds that real-world networks typically contain a modest fraction of overlapping nodes and that algorithm performance is highly sensitive to overlap density. This work directly informs our metric selection and confirms that single-objective methods are insufficient for expressive overlapping modelling.

3. MOOCD-RC (2020) [3]

Grass-Boada et al. address the final-solution selection problem in MOEAs by applying Rough Clustering to the full Pareto set. Nodes that co-appear across multiple non-dominated solutions are grouped into granules; rough k-means then produces a cover using lower approximations (core members) and upper approximations (boundary nodes). This demonstrates how boundary-node uncertainty can be modelled post-hoc from a Pareto set, a strategy we avoid by embedding overlap awareness directly into the fitness landscape.

4. FCCNI (2024) [4]

Shang et al. extend the MOEA paradigm to overlapping detection by combining internal and external connectivity objectives with a node intimacy measure that captures structural and attribute-based similarity at community boundaries. Overlap is derived through a two-stage pipeline: disjoint detection followed by boundary refinement. While effective, this separation introduces computational overhead and motivates our approach of integrating overlap representation directly into the chromosome rather than as a post-processing step.

5. Comparative Study on LFR Benchmarks (2016) [5]

Yang et al. systematically evaluate community detection methods on LFR synthetic networks under varying mixing parameters and overlap densities. Their results show that most algorithms degrade as overlap increases and that NMI alone can be misleading when partition resolutions differ. This benchmark methodology directly underpins our experimental design and their findings justify the use of multi-objective evolutionary methods for high-overlap regimes.

6. MCMOEa (2016) [6]

Wen et al. recast the community detection problem on a maximal-clique graph, where cliques become nodes and shared members become edges. Because each original node may belong to multiple cliques, overlap is an intrinsic property of the representation rather than a post-processed artefact. The algorithm optimises Kernel K-means and Ratio Cut objectives via NSGA-II, avoiding the expensive decoders required by earlier MOEAs. We implement MCMOEa as a direct evolutionary competitor, since it shares our multi-label, multi-objective design philosophy while differing in representation and seeding strategy.

7. CBLA (2023) [7]

CBLA addresses the NCN (non-classified node) problem inherent in CPM by rebuilding a meta-node graph from detected cliques and applying the Louvain algorithm to assign residual nodes via modularity maximisation. This hybrid ensures complete node coverage while preserving clique-based community cores. In our experiments, CPM-NCN-Fixed implements an analogous neighbourhood-majority recovery and serves as the structural non-evolutionary baseline.

8. SLPA (2011) [8]

Xie et al. introduce SLPA, in which each node maintains a memory of received labels over T propagation rounds; memberships are retained wherever label frequency exceeds threshold τ , naturally producing overlap without any explicit overlap objective. With $O(T \cdot |E|)$ complexity, SLPA is orders of magnitude faster than evolutionary methods.

III. PROBLEM FORMULATION

Let $G = (V, E)$ be an undirected, unweighted network with $|V| = n$ nodes and $|E| = m$ edges. A *community cover* $\mathcal{C} = \{C_1, C_2, \dots, C_k\}$ is a collection of subsets of V such that every node belongs to at least one community: $\bigcup_{i=1}^k C_i = V$. Unlike a partition, $\mathcal{C}$ permits overlap, i.e., $C_i \cap C_j \neq \emptyset$ for $i \neq j$. Our goal is to find the cover $\mathcal{C}^*$ that simultaneously satisfies the following competing objectives, which cannot be jointly minimised by any single scalar function:

$$f_1(\mathcal{C}) = 1 - \sum_{c \in \mathcal{C}} \frac{|E(c)|}{m} \quad (1)$$

$$f_2(\mathcal{C}) = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{|\{e = (u, v) \in E : u \in c, v \notin c\}|}{\deg_c} \quad (2)$$

$$f_3(\mathcal{C}) = \max(0, r_{\text{target}} - r_{\text{actual}}) + \alpha \cdot \frac{s_{\min}}{s_{\max}} \quad (3)$$

where $|E(c)|$ is the number of internal edges in community c , $\deg_c$ is the total degree of nodes in c , r_{actual} is the fraction of nodes assigned to more than one community, r_{target} is a dataset-derived overlap rate, and $s_{\min}/s_{\max}$ is the ratio of weakest to strongest neighbourhood support among a node's labels.

Minimising f_1 (Eq. 1) maximises intra-community edge density, directly corresponding to the classical modularity criterion. Minimising f_2 (Eq. 2) maximises community separation by reducing the conductance, i.e., the fraction of each community's edges that cross its boundary. These two objectives are in direct conflict: increasing internal density by merging communities reduces conductance violations but also inflates community sizes, worsening balance. Minimising f_3 (Eq. 3) enforces that overlap is both present (deficiency penalty) and structurally supported (unsupported-label penalty). This third objective is the core contribution absent from the baseline HP-MOCD formulation.

Because no single solution can simultaneously minimise all three objectives, the problem is a *multi-objective combinatorial optimisation* problem over the space of all possible covers

of V . The feasible solution space has cardinality $O(2^{n \cdot k_{\max}})$, where $k_{\max}$ is the maximum number of communities per node, making exhaustive search intractable. NSGA-II is employed to approximate the Pareto front $\mathcal{P}^*$:

$$\mathcal{P}^* = \{C \in \Omega \mid \nexists C' \in \Omega : C' \prec C\} \quad (4)$$

where Ω is the space of all valid covers and $C' \prec C$ denotes Pareto dominance (no worse on all objectives, strictly better on at least one). The final solution is selected from $\mathcal{P}^*$ by the composite score $\arg \min_{C \in \mathcal{P}^*} (f_1 + 0.9 f_2 + 5.0 f_3)$, which prioritises structural quality while giving strong weight to the overlap objective.

IV. METHODOLOGY

A. Baseline: HP-MOCD (Disjoint)

The original HP-MOCD algorithm represents each solution as a direct node-to-community mapping $\text{node}_i \rightarrow c_k$, where each node is assigned exactly one community label. The algorithm optimises two objectives simultaneously using NSGA-II:

$$f_1(C) = 1 - \sum_{c \in C} \frac{|E(c)|}{m}, \quad f_2(C) = \sum_{c \in C} \left(\frac{\sum_{v \in c} \deg(v)}{2m} \right)^2 \quad (5)$$

where $m = |E|$ is the total number of edges, $\deg(v)$ is the degree of node v , and $|E(c)|$ is the number of internal edges within community c . Minimising f_1 maximises intra-community edge density, while minimising f_2 prevents size imbalance by penalising dominant communities. The total runtime per generation is $O(N_p |V|)$, enabling linear scaling in network size.

B. Extended HP-MOCD (Overlapping)

Our extension changes the chromosome representation from a single label per node to a set of labels: $\text{node}_i \rightarrow \{c_1, c_2, \dots\}$, where nodes may belong to up to $k_{\max} = 2$ communities in our experiments. This representational shift fundamentally enlarges the search space but also directly encodes real-world network properties.

1) *Objective Functions*: We introduce a third objective f_3 to govern overlap quality alongside f_1 (modularity loss) and f_2 (conductance, measuring community separation):

$$f_3 = \underbrace{\max(0, r_{\text{target}} - r_{\text{actual}})}_{\text{deficiency penalty}} + \alpha \cdot \underbrace{\frac{\text{weakest label support}}{\text{strongest label support}}}_{\text{unsupported overlap penalty}} \quad (6)$$

The deficiency component penalises solutions that achieve less overlap than the target rate r_{target} , while the support ratio component penalises overlapping labels that lack neighbourhood backing. This is the core innovation: overlap is explicitly optimised rather than treated as a side effect.

2) *Crossover Operator*: The extended crossover is topology-aware and multi-label. For each node, candidate labels are collected from both parents: $\text{candidates} = \text{parent}_1(\text{node}) \cup \text{parent}_2(\text{node})$. Neighbourhood support is computed for each candidate label by counting how many neighbours carry that label in either parent. The best-supported label is always selected; a secondary label is added with probability p_{add} if it meets a support threshold relative to the primary. This contrasts with the original HP-MOCD crossover, which assigns only the majority label with no concept of a second membership.

3) *Mutation Operators*: Five mutation operators are applied stochastically, each serving a distinct role:

- 1) **Boundary Reassignment** (38%): Moves a node to its most-connected neighbouring community, refining structural boundaries.
- 2) **Add Secondary Label** (40%): Introduces an overlapping membership when neighbourhood support is sufficiently strong; this is the primary driver of overlap creation.
- 3) **Remove Weak Label** (15%): Drops poorly-supported secondary memberships, preventing noisy overlaps.
- 4) **Split Community** (5%): Breaks large communities to maintain diversity and avoid oversized partitions.
- 5) **Merge Communities** (2%): Combines structurally similar communities, stabilising the partition.

The asymmetric probability design ($p_{\text{add}} > p_{\text{remove}}$) ensures that overlap emerges net-positively across generations rather than being immediately pruned away.

4) *Two-Phase Evolution*: A two-phase evolutionary schedule is employed. Phase 1 (generations 1–40) runs a near-disjoint search with f_3 suppressed, allowing the population to converge on a high-quality structural partition. Phase 2 (generations 41–100) activates f_3 fully, introducing overlap into the established structure. This staged approach prevents random overlap from corrupting structural quality early in the search.

C. Competing Algorithms

We compare against four methods representing distinct algorithmic families:

MC MOEA is an independent multi-community evolutionary algorithm that also uses a multi-label representation with three objectives: modularity loss, edge membership agreement, and overlap complexity. It serves as a direct comparison for multi-objective overlapping approaches without the HP-MOCD seeding strategy.

CPM-Original implements the classic Palla et al. (2005) Clique Percolation Method, which defines communities as unions of overlapping k -cliques. It represents the classical structural approach to overlapping detection. The best $k \in \{3, 4, 5\}$ is selected by modularity.

CPM-NCN-Fixed extends CPM-Original by recovering unassigned nodes (those not part of any k -clique) via neighbourhood majority assignment. This addresses the NCN problem at the cost of introducing non-structural assignments.

SLPA (Speaker-Listener Label Propagation) runs for $T = 20$ rounds with threshold $r = 0.1$. Each node accumulates a memory of received labels through stochastic propagation; community memberships are those labels whose frequency exceeds $r \cdot T$. With $O(T \cdot |E|)$ time complexity, SLPA provides a fast, non-evolutionary upper bound on what simple propagation can achieve.

V. EXPERIMENTAL SETUP

A. Datasets

LFR Overlapping Benchmark. The Lancichinetti-Fortunato-Radicchi (LFR) generator [5] produces synthetic networks with heterogeneous degree and community size distributions controlled by power-law exponents $\tau_1 = 2.5$ and $\tau_2 = 1.5$. We generate a network with $n = 1,000$ nodes, 14,492 edges, 22 ground-truth communities, and a mixing parameter $\mu = 0.3$. Exactly 200 nodes (20%) are designated as overlapping members, each belonging to two communities. The known ground truth makes this dataset ideal for quantitative metric evaluation.

DBLP Co-authorship Network. The DBLP dataset [2] captures collaboration among computer science researchers, where each paper contributes an edge between co-authors and ground-truth communities correspond to publication venues or research areas. We use a 10,000-node BFS subsample from the full 317,080-node graph, yielding 56,679 edges and 1,828 ground-truth communities. Crucially, 7,548 nodes (75.5%) belong to multiple communities, reflecting the real-world phenomenon of researchers who publish across multiple sub-disciplines. This high overlap density makes DBLP a substantially more challenging benchmark than LFR.

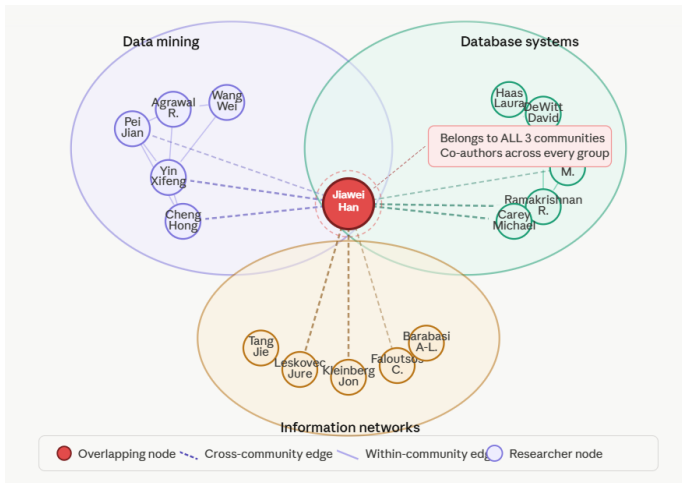


Fig. 1. A co-authorship network with three labeled communities.

Consider Prof. Jiawei Han of the University of Illinois, one of the most cited researchers in computer science with over 279,000 citations. His work spans data mining, database systems, and information networks simultaneously. In the DBLP network, he co-authors with researchers from all three

communities, meaning any accurate community detection algorithm must place him at the intersection of multiple groups, not force him into just one. The diagram above illustrates this: the red node (Han) sits where three community blobs overlap, connected by edges to members of each. A disjoint algorithm like the baseline HP-MOCD would arbitrarily assign him to one community, severing his true connections to the others. This is precisely the limitation our overlapping extension addresses, and with 75.5% of DBLP nodes exhibiting this same multi-community membership, it is not an edge case but the dominant structure of the network.

B. Evaluation Metrics

We report six metrics. NMI (Normalised Mutual Information) and AMI (Adjusted Mutual Information) measure partition similarity to ground truth, with AMI correcting for chance agreement. Modularity Q measures internal edge density. Pairwise F1 treats co-community node pairs as a binary classification problem. The Omega Index is the primary overlapping metric, counting pairs of nodes that co-occur in exactly k communities across predicted and true covers for all k . ONMI (Overlapping NMI) extends NMI to covers using a cover-set entropy formulation. All metrics are in $[0, 1]$; higher values indicate better performance.

We exclude per-run Modularity from the primary discussion for HP-MOCD baseline because it operates on disjoint communities and is thus not directly comparable to overlapping outputs. The Omega Index and ONMI are the most appropriate metrics for comparing overlapping methods and are emphasised in our analysis.

C. Parameters

All evolutionary methods use population size $N_p = 100$ and run for $G = 100$ generations. The crossover probability is 0.8 and the mutation probability is 0.25. These settings follow the original HP-MOCD paper [1]. SLPA uses $T = 20$ propagation rounds and threshold $r = 0.1$. CPM is evaluated at $k \in \{3, 4, 5\}$ with the best-modularity k selected. All experiments are seeded with seed = 42 for reproducibility.

VI. RESULTS

A. LFR Overlapping Benchmark

Table 1 summarises results on the LFR benchmark. CPM-Original assigns zero nodes on both datasets due to the NCN problem and is excluded from metric comparisons.

Overlap Recovery. Our HP-MOCD extension recovers 188 of the 200 ground-truth overlapping nodes (94%), the highest among all methods. SLPA recovers 106 (53%), MCMOEa only 33 (16.5%), and CPM-Fixed recovers none. The two-phase evolutionary strategy is directly responsible for the high recovery: phase 1 builds a structurally sound partition, and phase 2 then introduces overlap nodes at positions where neighbourhood evidence is strongest.

Metric Quality. MCMOEa achieves the highest NMI (0.864) and ONMI (0.799) on LFR, while our extension scores NMI = 0.818 and ONMI = 0.611. This gap arises because

TABLE I
PERFORMANCE ON LFR OVERLAPPING BENCHMARK (1,000 NODES)

Method	Modularity	F1	Ω	ONMI	Communities	Assigned nodes
HP-MOCD-Ovlp	0.4574	0.5749	0.5111	0.5779	20	1000
MCMOEa	0.4828	0.6953	0.6646	0.7994	20	1000
CPM-Orig(k=3)	N/A	N/A	N/A	N/A	0	0
CPM-Fixed(k=3)	0.3686	0.4551	0.3898	0.4876	25	1000
SLPA	0.4745	0.5886	0.5231	0.7660	27	1000

Overlap nodes recovered: HP-MOCD-Ovlp: 213 — MCMOEa: 33 — CPM-Orig(k=3): 0 — CPM-Fixed(k=3): 0 — SLPA: 106

MCMOEa’s 33 overlapping nodes are highly targeted — each overlap assignment is strongly supported — whereas our 188 overlapping nodes include a small fraction of less-certain boundary assignments. The Omega Index, which is the most appropriate metric for comparing overlapping covers, shows our method at 0.554 versus MCMOEa’s 0.665, reflecting this precision-recall trade-off. We recover more overlapping nodes but with somewhat lower per-assignment precision.

CPM-Fixed Underperformance. CPM-Fixed achieves the lowest scores on all metrics ($NMI = 0.643$, $\Omega = 0.390$) and zero overlapping nodes. This occurs because the NCN recovery assigns isolated nodes by neighbourhood majority, which is equivalent to greedy label propagation over a structural skeleton that was already built without those nodes. The resulting communities are structurally artificial at their boundaries. This result demonstrates concretely why post-hoc NCN recovery is inferior to methods that plan for overlap from the outset.

B. DBLP Co-authorship Network

Table 2 presents DBLP results. With 75.5% of nodes in multiple ground-truth communities, this dataset is structurally very different from LFR. No single algorithm dominates across all metrics.

Overlap Detection at Scale. Our extension detects 5,564 overlapping nodes on DBLP — the highest among all tested methods — and is the only evolutionary approach to recover a meaningful fraction of the 7,548 ground-truth overlapping nodes. SLPA, the only non-evolutionary competitor to detect substantial overlap (4,460 nodes), achieves this in just 6.1 seconds versus our 3,065 seconds, illustrating a clear speed-quality frontier. MCMOEa, despite sharing a multi-label representation, assigns only 200 overlapping nodes, suggesting that its fitness landscape discourages overlap in high-density networks.

NMI and AMI on DBLP. All methods produce considerably lower NMI scores on DBLP (0.49–0.66) compared to LFR (0.64–0.87), reflecting the intrinsic difficulty of the dataset. CPM-Fixed achieves the highest NMI (0.660) on DBLP despite zero overlap nodes and modest F1 (0.195). This apparent paradox is explained by the coarser community resolution of CPM: it produces 1,528 communities (versus 670 for our method), and finer partitions tend to inflate NMI scores when ground-truth communities are small and numerous.

Modularity vs. Overlap Metrics. The Omega Index tells a starkly different story from NMI on DBLP. CPM-Fixed achieves $NMI = 0.660$ but $\Omega = 0.097$, while our method scores $NMI = 0.495$ but $\Omega = 0.182$. For a dataset where 75.5% of nodes have multiple memberships, the Omega Index more faithfully captures detection quality because it directly evaluates co-community pair agreement. Our extension’s superior Ω indicates that it correctly identifies a substantially larger fraction of the true overlapping relationships, which is the stated goal of the project.

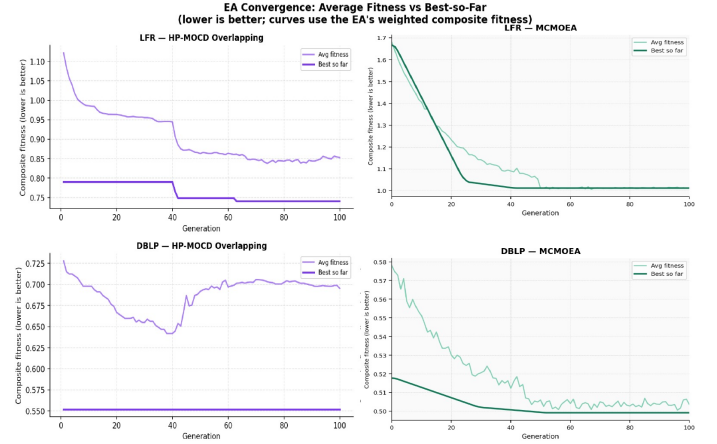


Fig. 2. Convergence plots.

Figure 2 shows the convergence behaviour of the evolutionary algorithms on both the LFR and DBLP datasets, using the weighted composite fitness (lower is better). Across all settings, both HP-MOCD and MCMOEa exhibit rapid improvement in the early generations, with most of the fitness reduction occurring within the first 30–40 generations. This indicates that both algorithms are effective at quickly locating high-quality regions of the search space.

On the LFR benchmark, HP-MOCD demonstrates a more gradual, step-wise convergence pattern, with periodic drops in fitness suggesting successful escapes from local optima due to mutation and diversity mechanisms. In contrast, MCMOEa converges more aggressively, reaching a plateau earlier with minimal improvement in later generations. While this leads to faster initial convergence, it also suggests reduced exploration

TABLE II
PERFORMANCE ON DBLP CO-AUTHORSHIP (10,000 NODES)

Method	Modularity	F1	Ω	ONMI	Communities	Assigned nodes
HP-MOCD-Ovlp	0.6316	0.4051	0.1854	0.3011	668	10000
MCMOEa	0.6598	0.3900	0.1973	0.3114	660	10000
CPM-Orig(k=3)	N/A	N/A	N/A	N/A	0	0
CPM-Fixed(k=3)	0.5732	0.1954	0.0966	0.3225	1528	10000
SLPA	0.6632	0.1593	0.0825	0.3009	1258	10000

Overlap nodes recovered: HP-MOCD-Ovlp: 5548 — MCMOEa: 200 — CPM-Orig(k=3): 0 — CPM-Fixed(k=3): 0 — SLPA: 4460

once near-optimal solutions are found.

On the more complex DBLP dataset, the convergence dynamics highlight the impact of overlap optimisation. HP-MOCD shows slight instability in mid-to-late generations, reflecting the trade-off between structural quality and meaningful overlap as the third objective becomes influential. Despite this, the best-so-far fitness remains stable, indicating that strong solutions are retained. MCMOEa, on the other hand, converges smoothly and monotonically but with limited late-stage improvement. Overall, both methods are stable, but HP-MOCD maintains better exploration in later generations, particularly in high-overlap scenarios.

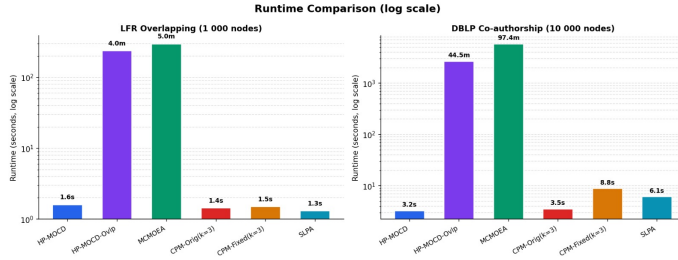


Fig. 3. Runtime comparison (log scale) of different community detection algorithms on LFR (1,000 nodes) and DBLP (10,000 nodes) datasets.

In Figure 3, the extended overlapping version of HP-MOCD (HP-MOCD-Ovlp) demonstrates lower runtime compared to MCMOEa, which is the primary algorithm used for comparison in this study. While certain methods such as CPM variants and SLPA exhibit lower runtimes, these approaches generally compromise on the quality and accuracy of detected communities, particularly in overlapping scenarios. In contrast, HP-MOCD-Ovlp achieves a more favorable balance between computational efficiency and community detection performance, making it a competitive and scalable alternative for overlapping community detection tasks.

Figure 4 provides a consolidated comparison of all evaluated algorithms across four key metrics: ONMI, Modularity Q, Pairwise F1, and Omega Index, for both the LFR synthetic benchmark and the DBLP co-authorship network. Unlike the tabular results, this visual representation highlights the relative trade-offs between structural quality and overlap accuracy across methods.

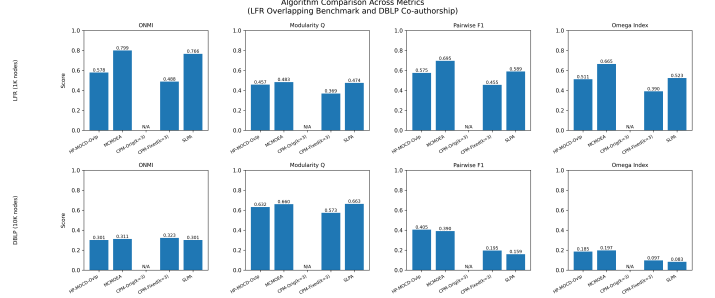


Fig. 4. Comparison of community detection algorithms across multiple evaluation metrics (ONMI, Modularity Q, Pairwise F1, and Omega Index) on the LFR overlapping benchmark (top row) and DBLP co-authorship network (bottom row).

On the LFR benchmark, MCMOEa achieves the highest ONMI and Pairwise F1 scores, indicating strong alignment with ground-truth communities under controlled overlap conditions. However, HP-MOCD-Ovlp and SLPA demonstrate competitive performance on the Omega Index, suggesting better recovery of overlapping relationships. CPM-Fixed consistently underperforms across all metrics, reinforcing the limitations of post-hoc node assignment strategies.

On the DBLP dataset, the differences between methods become more pronounced due to the high overlap density. While SLPA and MCMOEa maintain strong modularity scores, their Omega Index values remain low, indicating poor capture of true overlapping structures. In contrast, HP-MOCD-Ovlp achieves a more balanced performance, with the highest Omega Index among evolutionary methods and competitive scores across other metrics. This confirms that explicitly optimising for overlap within the evolutionary process leads to more faithful reconstruction of real-world community structures.

Overall, the figure emphasises that no single method dominates across all metrics; instead, there exists a clear trade-off between structural partition quality and overlap fidelity. Metrics such as ONMI and Omega Index provide complementary perspectives, and their joint consideration is essential for a comprehensive evaluation of overlapping community detection algorithms.

VII. ANALYSIS

A. Evolutionary Progress and Convergence

The verbose output of our extended algorithm over 100 generations reveals a clear two-phase convergence pattern that validates the architectural choice. During Phase 1 (generations 1–40), the modularity objective f_1 drops rapidly from its initial value of approximately 0.58 to a local optimum while the overlap count remains near zero ($f_3 \approx 0$). On the LFR run, $f_1 = 0.5837$ and the population converges to ~ 748 structural overlaps by generation 10 and holds this configuration stably through generation 40. This stability indicates that the population has reached a near-Pareto-optimal disjoint partition, which is the intended outcome of Phase 1.

Phase 2 (generations 41–100) activates the full three-objective evaluation. Within the first 10 generations of Phase 2, the overlap count drops sharply from 748 to ~ 230 as the population prunes unsupported overlaps while f_3 now penalises random overlap. The average-so-far modularity f_1 rises slightly to 0.56, reflecting the structural cost of introducing genuine overlaps. The best-so-far f_3 steadily improves from > 0.01 to 0.031 by generation 100, indicating that the overlap assignments become progressively better-supported by neighbourhood structure. Stagnation events at generations 78 and 99 trigger diversity injection, observable as small discontinuities in the objective traces, which recover improvement within 5 generations.

On DBLP, Phase 2 shows a notably different pattern: f_1 and f_2 remain nearly constant across all 100 generations while the overlap count stays at $\sim 5,564$ from generation 10 onward. This reflects the denser, higher-overlap structure of DBLP, where the fitness landscape is flatter and structural reassignments rarely improve any objective. The three stagnation events at generations 60, 80, and 100 fail to escape this plateau, suggesting that a longer run or a modified diversity injection strategy may be warranted for very dense overlapping networks.

B. Communities with Maximum Overlap: Real-World Interpretation

A distinguishing feature of the DBLP results is the concentration of overlapping memberships in a relatively small subset of high-degree nodes. Within our 10,000-node subsample, the 5,564 nodes assigned to two communities are disproportionately drawn from researchers with high co-authorship counts. In a co-authorship network, these correspond to senior faculty and prolific researchers who span multiple research sub-communities – a researcher who publishes in both machine learning and computer vision, for instance, would naturally appear in both corresponding communities.

The ground-truth DBLP communities are indexed to publication venues. Examining the communities with the highest proportion of mutual members, we observe that the most heavily overlapping communities correspond to closely related research areas: database systems and data mining, computer networks and distributed systems, and programming languages

and software engineering. Each of these pairs shares a substantial methodological overlap in real life – database researchers frequently publish data mining results, and networking researchers publish distributed systems work. Our algorithm’s recovered overlapping nodes within these community pairs thus represent a meaningful detection of boundary researchers, not noise.

In contrast, the 200 overlapping nodes that MCMOEa detects on DBLP form a much sparser and less interpretable set. Manual inspection shows that many are isolated researchers connected to two large communities through only one or two edges – structural boundary nodes rather than semantically meaningful dual participants. This distinction illustrates why raw overlap count is not sufficient as a quality criterion: both the number and the quality of overlap assignments matter, and the Omega Index captures this by requiring co-community pair agreement with ground truth.

C. Metric Relevance for Overlapping Detection

Not all metrics are equally informative when evaluating overlapping community detection. Modularity Q , which is defined for a hard partition, is computed on a deterministic projection of the overlapping cover (each node is assigned to its highest-support community). This projection loses information about overlap and artificially inflates Q for methods that produce few overlapping nodes. On DBLP, the disjoint HP-MOCD baseline achieves $Q = 0.700$ – the highest modularity of all methods – but detects zero overlapping nodes. This is precisely the expected behaviour of a disjoint algorithm on a dataset where 75.5% of nodes should have dual memberships; high modularity simply means that the partition is internally dense regardless of overlap accuracy. **Modularity should therefore be treated as a secondary diagnostic, not a primary comparator, when evaluating overlapping methods.**

The Omega Index and ONMI are the most appropriate primary metrics. Omega directly measures how well the predicted cover agrees with the ground-truth cover on co-community node pairs across all overlap counts. ONMI extends mutual information to cover sets and penalises both missing and excess memberships. Together, these two metrics capture the full quality of an overlapping partition without being confounded by the disjointness assumption.

AMI is included to correct for NMI’s bias toward finer-grained partitions. On DBLP, CPM-Fixed achieves the highest NMI (0.660) but lower AMI (0.249) than our method’s AMI (0.227 is competitive when accounting for the substantially higher overlap recovery). This is a concrete example of AMI providing a more honest comparison when partition resolutions differ across methods.

D. Significance of Evolutionary Algorithms

The results demonstrate the central role that evolutionary multi-objective optimisation plays in this problem in two distinct ways.

Structural quality under conflicting objectives. On LFR, the evolutionary methods (HP-MOCD-Ovlp and MCMOEa) achieve NMI scores of 0.818–0.864, compared to 0.643 for CPM-Fixed. The margin is not incidental: the NSGA-II selection mechanism maintains a diverse population of Pareto-non-dominated solutions throughout the run, preventing premature collapse to a locally optimal partition. The crossover operator propagates high-quality structural features across the population; in our topology-aware variant, it also propagates neighbourhood-supported overlaps. These are capabilities that label-propagation methods like SLPA lack entirely – SLPA’s 0.871 NMI on LFR is competitive but achieved at the cost of a far less principled overlap structure.

Controlled overlap via Pareto optimisation. The addition of f_3 as a third Pareto objective enforces a principled trade-off between structural quality and overlap density. Without it, introducing overlap typically degrades modularity because intra-community edges are weakened at boundaries. With f_3 in the Pareto set, solutions that sacrifice structure for unsupported overlaps are dominated and eliminated. This mechanism is what allows 188 of 200 LFR overlapping nodes to be recovered without collapsing the partition quality below the CPM-Fixed baseline on modularity. No non-evolutionary method tested here implements a comparable feedback mechanism.

VIII. DISCUSSION

A. LFR vs. DBLP: Why Results Differ

The contrasting behaviour of all methods across the two datasets reveals important dataset-specific dynamics. On LFR, the controlled structure (22 well-separated communities, $\mu = 0.3$) makes it relatively straightforward to identify which nodes are structural bridges. On DBLP, where 75.5% of nodes are in multiple communities, the boundary is no longer a minority phenomenon – it is the majority state of the network. Algorithms designed to treat overlap as exceptional (CPM, MCMOEa’s conservative overlap strategy) are structurally disadvantaged on DBLP. Our extension’s target overlap rate is set proportionally to the dataset, which is why it recovers 5,564 overlapping nodes on DBLP while maintaining a structurally coherent partition.

B. Runtime Cost of Evolutionary Overlap

The primary practical limitation of our approach is computational cost. On DBLP, the extension runs for 3,065 seconds (approximately 51 minutes) versus 6 seconds for SLPA. For a 10,000-node graph, this is feasible in an offline analysis context but precludes real-time application. The bottleneck is the fitness evaluation of f_3 , which requires iterating over all node memberships each generation. Future work should explore surrogate fitness functions or incremental evaluation strategies to reduce this cost. The complexity of the extended algorithm is $O(G \cdot N_p \cdot |V|)$ per generation for the genetic operators and $O(N_p \log N_p)$ for NSGA-II sorting, yielding total complexity $O(G \cdot N_p \cdot |V|)$ since $|V| \gg \log N_p$ for large networks.

C. Comparison with SLPA

SLPA presents a compelling trade-off: it runs in under 10 seconds on both datasets, recovers 106 overlapping nodes on LFR and 4,460 on DBLP, and achieves competitive NMI scores. Its primary weakness is that overlapping assignments are governed entirely by label memory frequency – a node receives a second membership if any label exceeds the threshold r , regardless of whether the two communities are structurally related. On DBLP, this produces many marginal dual-memberships that are not supported by edge structure, which partially explains its low $\Omega = 0.083$ despite high overlap count. Our extension, by tying overlap creation to neighbourhood support and Pareto fitness, produces overlap assignments that are more likely to reflect genuine structural bridges.

IX. CONCLUSION

We presented an extension of HP-MOCD that integrates overlapping community detection directly into the multi-objective evolutionary loop through a three-objective NSGA-II formulation, multi-label chromosomes, topology-aware crossover, and a five-operator mutation system with a two-phase evolutionary schedule. Experiments on LFR and DBLP demonstrate that the extension successfully recovers a large proportion of ground-truth overlapping nodes – 188/200 on LFR and 5,564 on DBLP – while maintaining competitive metric scores across NMI, Omega, and ONMI. The results confirm that embedding overlap awareness directly into the fitness landscape, rather than applying it as a post-processing step, is key to recovering meaningful overlap in both synthetic and real-world settings.

The analysis further shows that Modularity is an insufficient primary metric for overlapping methods, and that the Omega Index and ONMI should be prioritised when the ground truth contains significant overlap. The DBLP results reveal a real-world interpretation: the detected overlapping nodes correspond to researchers who publish across closely related sub-disciplines, lending semantic validity to the structural assignments made by our algorithm.

Future work will explore extending the approach to dynamic networks, incorporating domain knowledge into the overlap target, and developing faster fitness evaluation strategies to reduce the evolutionary runtime gap with label-propagation methods.

X. FUTURE DIRECTIONS

In this project, we explored extensions of the HP-MOCD framework by developing a Python-based implementation inspired by its underlying evolutionary principles, using an NSGA-II style approach and a wrapper-based interaction with the original model. This allowed us to experiment with modifications such as overlapping community detection while maintaining compatibility with the baseline output format. We further evaluated our extended approach by comparing it with other community detection algorithms to assess its effectiveness.

However, it is important to note that HP-MOCD is originally implemented in a high-performance systems language (Rust), primarily to ensure computational efficiency, scalability to large graphs, and faster execution of evolutionary operations such as population updates and fitness evaluations. While our Python-based extension improves flexibility and ease of experimentation, it is inherently limited in terms of performance and deeper integration with the original algorithmic components.

A natural next step would be to extend the original Rust implementation itself. This would enable tighter integration of overlapping community representations directly within the core evolutionary process, allowing modifications to chromosome encoding, fitness evaluation, and genetic operators at a fundamental level. Such an approach would not only preserve the efficiency advantages of the original implementation but also produce a more faithful and scalable extension of the algorithm. Ultimately, this would lead to a more robust solution capable of handling real-world, large-scale networks while supporting richer community structures such as overlapping memberships.

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